Report XI

Estimation of nonlinear fitting parameters with Conjugate Gradients and Golden Section methods

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1. Introduction

In this project, nonlinear fitting of tabulated data was considered. The aim was to approximate a set of points:

$$(x_i, y_i, w_i)$$

where:

$$x_i$$

are approximation nodes,

$$y_i$$

are measured values,

$$w_i$$

are weights.

The fitting function depends on a vector of parameters:

$$\vec{a} = [a_0,\ a_1,\ a_2,\ a_3]$$

The function used in this task was:

$$p(x, \vec{a}) = a_0 \arctan\,(a_1 x + a_2) + a_3$$

This function contains both linear and nonlinear parameters. Parameters $a_0$and $a_3$are linear, while a_1 and a_2 are nonlinear. Therefore, the fitting problem cannot be solved only as a simple linear approximation problem.

To find the best parameters, the mean-square approximation method was used. The objective function was defined as:

$$f(\vec{a}) = \sum_{i=0}^{n-1} w_i \,(y_i - p(x_i, \vec{a}))^2$$

The task was to minimize this function. A smaller value of $f(\vec{a})$ means that the fitted function is closer to the experimental data.

The minimization was performed using the Conjugate Gradients method. In each iteration, the gradient of the objective function was calculated, and the search direction was updated. The step length in this direction was found using the Golden Section method.

Two variants of the Conjugate Gradients method were analyzed:

$$\gamma_{FR}$$

Fletcher-Reeves formula,

$$\gamma_{PR}$$

Polak-Ribiere formula.

The main aim of the exercise was to compare how these two formulas influence the convergence of the method for a nonlinear fitting problem. The instruction required testing both Fletcher-Reeves and Polak-Ribière formulas and comparing the obtained fitting functions with the exact one.

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2. Description of the Numerical Task

The approximation interval was:

$$x \in [-10, 20]$$

The number of nodes was:

$$n = 100$$

The nodes were distributed uniformly:

$$\Delta = \frac{x_{max} - x_{min}}{n - 1}$$
$$x_i = x_{min} + \Delta i$$

The exact parameters used to generate the data were:

$$a_0 = 0.5$$
$$a_1 = 10$$
$$a_2 = -30$$
$$a_3 = 0.7$$

The experimental data were generated from the exact function with added random noise:

$$y_i = p(x_i, \vec{a}) + \delta_i$$

where:

$$\delta_i = 0.1\left(\frac{rand()}{RAND\_MAX} - 0.5\right)$$

The weights were constant:

$$w_i = 1.0$$

The initial vector for the minimization was:

$$\vec{a}^{(0)} = [1, 1, 1, 1]$$

The stopping condition for the Conjugate Gradients method was:

$$|| \vec{a}_{k+1} - \vec{a}_k ||_2 < 10^{-4}$$

The stopping condition for the Golden Section method was:

$$\lambda_B - \lambda_A < 10^{-5}$$

The maximum number of iterations was:

$$K_{max} = 1000$$

for the Conjugate Gradients method and:

$$L_{max} = 100$$

for the Golden Section method.

The task was to perform the minimization for both Fletcher-Reeves and Polak-Ribiere formulas. Then, the obtained functions were compared with the exact fitting function and with the generated experimental data. The practical part of the instruction also asks to check the influence of a larger initial Golden Section range and different starting values of the vector $\vec{a}$.

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3. Numerical Method

The objective function was minimized using the Conjugate Gradients method. The gradient of the objective function was calculated from:

$$\nabla f(\vec{a}) = \sum_{i=0}^{n-1} (-2)\, w_i (y_i - p(x_i, \vec{a})) \left[\frac{\partial p}{\partial a_0}, \frac{\partial p}{\partial a_1}, \frac{\partial p}{\partial a_2}, \frac{\partial p}{\partial a_3}\right]$$

The derivatives of the fitting function were:

$$\frac{\partial p}{\partial a_0} = \arctan\,(a_1 x + a_2)$$

$$\frac{\partial p}{\partial a_1} = \frac{a_0 x}{(a_1 x + a_2)^2 + 1}$$

$$\frac{\partial p}{\partial a_2} = \frac{a_0}{(a_1 x + a_2)^2 + 1}$$

$$\frac{\partial p}{\partial a_3} = 1$$

In the first iteration, the search direction was equal to the negative gradient:

$$\vec{u}_0 = \vec{g}_0$$

where:

$$\vec{g}_k = -\nabla f(\vec{a}_k)$$

In the next iterations, the direction was updated as:

$$\vec{u}_k = \vec{g}_k + \gamma_k \vec{u}_{k-1}$$

For Fletcher-Reeves formula:

$$\gamma_{FR} = \frac{\vec{g}_k^T \vec{g}_k}{\vec{g}_{k-1}^T \vec{g}_{k-1}}$$

For Polak-Ribière formula:

$$\gamma_{PR} = \frac{(\vec{g}_k - \vec{g}_{k-1})^T \vec{g}_k}{\vec{g}_{k-1}^T \vec{g}_{k-1}}$$

After selecting the direction, the Golden Section method was used to find the step length:

$$\lambda = \arg \min_{\lambda} f(\vec{a} + \lambda \vec{u})$$

The vector of parameters was then updated:

$$\vec{a}_{k+1} = \vec{a}_k + \lambda \vec{u}_k$$

The algorithm was repeated until the stopping condition was satisfied or the maximum number of iterations was reached.

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4. Implementation

The program was written in C++ and followed the algorithm given in the instruction file. It did not use any numerical libraries.

Main steps of the implementation:

1. Data generation

- 100 equidistant nodes were generated on the interval $[-10, 20]$,
- exact function values were calculated using the given parameters,
- random noise was added to simulate experimental measurements,
- weights were set to $w_i = 1.0$,
- generated data were saved to a text file.

2. Fitting function

- function $p(x, \vec{a})$was calculated as:

$$p(x, \vec{a}) = a_0 \arctan(a_1 x + a_2) + a_3$$

3. Objective function

- the value of $f(\vec{a})$ was calculated as the sum of squared differences,
- all nodes and weights were included in the summation,
- this value was minimized by the algorithm.

4. Gradient calculation

- derivatives with respect to all four parameters were calculated,
- the gradient vector was computed from the formula given in the instruction file,
- the negative gradient was used as the descent direction.

5. Golden Section method

- the Golden Section method was used in every Conjugate Gradients iteration,
- it searched for the best value of λ,
- the initial interval was:

$$\lambda_A = 0$$
$$\lambda_B = 0.1$$

- the method stopped when the interval length was smaller than 10^{-5}.

6. Conjugate Gradients method

- calculations were performed for Fletcher-Reeves formula,
- calculations were performed for Polak-Ribière formula,
- after each iteration, parameters were updated,
- values of $f(\vec{a})$, λ, s, and parameters were saved to text files.

7. Data export

- data were saved to text files for plotting

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5. Results

The following cases were analyzed:

- Fletcher-Reeves formula,
- Polak-Ribière formula.

The generated experimental data were noisy, but the main shape of the function was clearly visible. The function had a rapid transition near:

$$x \approx 3$$

For smaller values of x, the function stayed close to the lower level. For larger values of x, it approached the upper level.

Case 1: Fletcher-Reeves formula

The Fletcher-Reeves method started correctly and the value of the objective function decreased during the first iterations.

The first value was approximately:

$$f(\vec{a}) = 19.2831$$

After several iterations, the function decreased, but later the method became unstable. The parameters started to move away from the correct solution, and the value of the objective function increased again.

In later iterations, the values became very large and finally the program produced:

$$inf$$

and

$$nan$$

This means that the Fletcher-Reeves formula did not converge correctly for this case.

Case 2: Polak-Ribière formula

The Polak-Ribière method gave much better results. The value of the objective function decreased from:

$$f(\vec{a}) = 19.2831$$

to approximately:

$$f(\vec{a}) = 0.0868$$

The obtained final parameters were approximately:

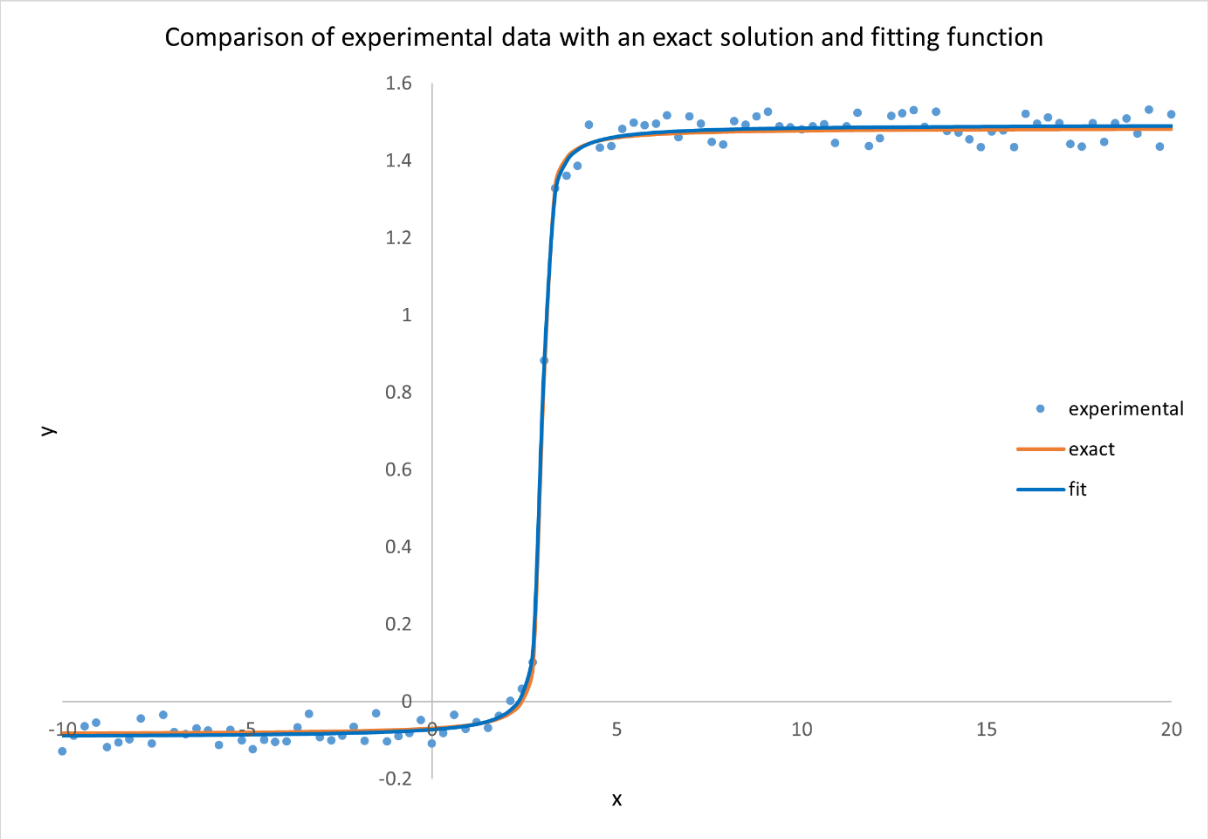
$$\begin{aligned} a_0 &= 0.5009 \\ a_1 &= 9.75 \\ a_2 &= -29.21 \\ a_3 &= 0.695 \end{aligned}$$

The exact values were:

$$a_0 = 0.5$$
$$a_1 = 10$$
$$a_2 = -30$$
$$a_3 = 0.7$$

The fitted parameters are close to the exact ones. The largest differences appear for a_1 and a_2, which control the position and steepness of the transition. Despite this, the fitted function follows the exact function and experimental data very well.

Figure 1. Fitting result obtained using Polak-Ribière formula.



Main observations:

- the objective function decreased to a small value,
- the fitted function was close to the exact function,
- the parameters were reconstructed correctly,
- the method was more stable than Fletcher-Reeves,
- the approximation was especially good outside the noisy transition region.

The Polak-Ribière formula was therefore more efficient in this task.

Additional test: larger Golden Section interval

An additional test was considered for a larger upper bound of the Golden Section interval:

$$\lambda_B = 1.0$$

while:

$$\lambda_A = 0$$

was left unchanged.

For this larger interval, the minimization became less stable. The algorithm could choose much larger steps in the search direction. In a nonlinear problem, such steps may move the parameters too far from the minimum.

Main observations:

- larger λ_B gives less control over the update,
- the method becomes more sensitive to the search direction,
- convergence may become worse,
- in some cases the algorithm may diverge.

Additional test: larger initial vector

The influence of a larger initial vector was also considered. When the initial parameters are much larger than:

$$[1, 1, 1, 1]$$

the starting point is farther from the correct minimum.

In this case, the gradient points into a direction which is not useful for finding the minimum. The values of the objective function can increase, and the algorithm tends to fail the converge.

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6. Conclusions

The Conjugate Gradients method with Golden Section line search can be used for nonlinear fitting problems.

The following observations were made:

- the fitting function was reconstructed from noisy experimental data,
- the objective function measured the difference between data and approximation,
- the Golden Section method was used to find the step length in every iteration,
- Fletcher-Reeves formula was unstable in this task,
- Fletcher-Reeves method diverged and produced invalid values,
- Polak-Ribière formula gave much better results,
- Polak-Ribière method reconstructed the exact parameters quite well,
- the final Polak-Ribière curve was close to the exact function,
- the method was sensitive to the initial vector,
- larger Golden Section interval could make the algorithm unstable.

The best result was obtained for the Polak-Ribière formula. The fitted parameters were close to the exact values and the final curve followed the experimental data well.

The Fletcher-Reeves formula did not work well for this case. It initially improved the result, but later the algorithm became unstable and diverged.

Overall, the exercise shows that the Conjugate Gradients method can solve nonlinear fitting problems, but its efficiency depends strongly on the formula used for γ, the line-search interval and the initial parameter vector.